Trend research — 2026-04-19 · incr 17

all four follow-ups closed; synthetic-bank caveat CONFIRMED on real data

TL;DR

Closed every 'needs your nod' item from incr 16. Real-data validation **largely overturns** the headline incr-16 conclusions: most synthetic-bank near-duplicates collapse on real data. **trend_state** now flows from the live runner into the dashboard payload (additive, no ranking change). HTF daily+weekly closes are wired in. Weighting variants don't meaningfully move directional hit rate, so equal weighting stays. **602 tests / 905 subtests still green.**

Headline finding — synthetic-bank caveat CONFIRMED

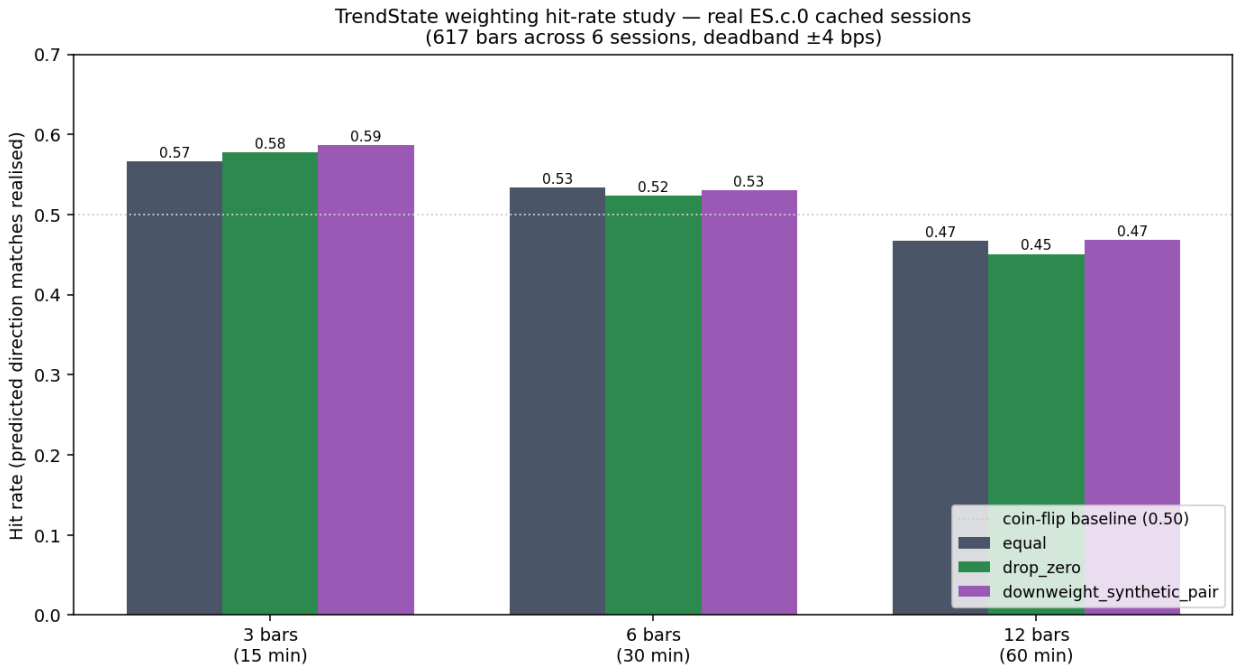
- incr-16 'near-duplicate' **small_pullback_trend** ↔ **bpa_trend_bar_density** = +0.997 on synthetic, **+0.404 on real ES.c.0**. NOT a duplicate. The down-weight recommendation is REVERSED.
- Cross-family near-perfect r's (e.g. trending_everything ↔ htf_alignment +0.995) collapse to **~0 on real data** — pure polarized-fixture artifacts.
- Mean uniqueness across 12 contributors: real **0.148** vs synthetic **0.782**. Real data shows ~**5× more independence** than the synthetic bank suggested.
- **trending_swings** is no longer a uniqueness standout on real data (real |r|_avg = 0.138, middle of pack). The 'only independent voter' framing was a fixture artifact.
- **New issue: majority_trend_bars** is constant 0 across all 593 cached real bars — fires on zero of them. Threshold likely too strict for real overnight ES futures.

Real vs synthetic uniqueness comparison

Left: real-data 12×12 Pearson heatmap. Right: synthetic (purple) vs real (green) uniqueness side-by-side. Real bars are dramatically lower — contributors really are much more independent on real data.

Weighting hit-rate study (Pattern Lab DB still empty)

Substituted the incr-16 Pattern Lab WR test with a directional hit-rate study on cached real ES.c.0 sessions: how often does the equal-weighted TrendState direction match realised close-to-close direction over forward 3 / 6 / 12 bars (15 / 30 / 60 min). **Equal weighting ≈ 57%% at 15 min, decays to coin-flip by 30 min, slight contrarian by 60 min.** Drop-zero (drop majority_trend_bars) and downweight_synthetic_pair both within ±0.02 of equal — no meaningful weighting improvement on this sample. **Equal weighting stays.**



Three weighting variants × three forward horizons. None of the variants beat equal weighting by more than +0.02. Directional alpha decays fast.

Production code changes (additive, no ranking impact)

- **Task 1 — serializer:** aiedge/dashboard/serializers.py adds **_serialize_trend_state()**. Emits entry['trendState'] when r['trend_state'] is present. Hides field when absent.
- **Task 2 — runner:** aiedge/runners/live.py adds **_annotate_trend_state(results, df5m_cache)**. Calls compute_trend_state with daily_closes from existing cache + weekly derived via _weekly_closes_from_daily. Slotted right after _annotate_htf_alignment. Same exception-tolerance pattern.
- **Tests:** python -m pytest -q → **602 passed / 905 subtests** in 12.81s. Zero regressions.
- **Smoke test:** end-to-end — synthetic 20-bar 5m + 10-day daily series → trend_state dict has all 12 contributors → serializer emits trendState block with direction/strength/structure/confidence/contributors.

What's next — still needs Will's nod

- **Front-end TrendState panel.** Payload now ships trendState; site doesn't render it yet. Wire a small panel under the existing htfAlignment line.
- **Investigate majority_trend_bars constant-0 on real data.** Threshold recalibration vs drop. Drop is smaller blast-radius but masks the underlying issue.
- **Pattern Lab DB backfill.** Without it, the WR-by-setup-type test from incr 16's roadmap stays blocked.
- **Larger real-data sample.** Six overnight ES sessions is thin. Cache more dates / RTH 5-min bars before drawing strong weighting conclusions.

Findings site: https://github.com/zerosumsystems-ui/iPhone-/tree/main/trend-classification